

Figure 10.2 The grid layout has automatically adjusted to the size of our new dataset. Notice that our new elements are above the old elements, but our layout has changed in size from a 4 x 4 grid to a 5 x 5 grid, causing the old elements to move to their newly calculated position.

The results in figure 10.2 show snapshots of the animation from the old position to the new position of the circles.

Calculating a scale based on what you know to be the grid size results in an inefficient piece of code. That wouldn't be useful if someone put in a different dataset. Instead, when designing layouts you'll want to provide functionality so that the layout size can be declared, and then any adjustments necessary happen within the code of the layout that processes data. To do this, we need to add a scoped size variable and then add a function to our `processGrid` function to allow the user to change that size variable. Sending a variable sets the value, and sending no variable returns the value. We achieve this by checking for the presence of arguments using the `arguments` object in native JavaScript. The updated function is shown in the following listing.

Listing 10.5 `d3.layout.grid` with size functionality

```
d3.layout.grid = function() {
var gridSize = [0,10];
     var gridXScale = d3.scale.linear();
var gridYScale = d3.scale.linear();
function processGrid(data) {
        var rows = Math.ceil(Math.sqrt(data.length));
           var columns = rows;

        gridXScale.domain([1,columns]).range([0,gridSize[0]]);
           gridYScale.domain([1,rows]).range([0,gridSize[1]]);
           var cell = 0;

        for (var i = 1; i <= rows; i++) {
          for (var j = 1; j <= columns; j++) {
            if (data[cell]) {
              data[cell].x = gridXScale(j);
              data[cell].y = gridYScale(i);
              cell++;
            }
            else {
              break;
            }
          }
        }
```

Initializes the variable with a default value

Creates two scales but doesn't define their range or domain

Defines the range and domain each time the layout is called

Applies the scaled values as x and y